

নিচে ScopeBridge-এর final Project Requirements দিলাম—আগের পুরো workflow অনুযায়ী, এবং backend হিসেবে Node.js + Express.js + TypeScript ধরে। Strategic plan-এর core concept, 5 AI features, 13 core features এবং architecture অনুসরণ করা হয়েছে। Pasted markdown

ScopeBridge — Project Requirements

1. Project Objective

ScopeBridge হবে একটি **client-to-development workflow and scope management platform**, যেখানে software agency এবং client-এর মধ্যে project requirement সংগ্রহ থেকে শুরু করে requirement clarification, AI analysis, scope approval, development task creation এবং পরবর্তীতে scope change management—পুরো workflow একটি centralized system-এর মাধ্যমে পরিচালনা করা যাবে।

মূল লক্ষ্য:

> Client কী চেয়েছে → কী requirement তৈরি হয়েছে → কী clarify হয়েছে → কী approve হয়েছে → কোন task তৈরি হয়েছে → পরে কী change চাওয়া হয়েছে → change-টি scope-এর মধ্যে নাকি বাইরে — পুরো বিষয়টি traceable রাখা।

ScopeBridge-এর মূল উদ্দেশ্য হলো **scope creep কমানো এবং software agency-এর revenue protect করা**।

2. Target Users

Primary Users

- Small Software Agencies
- Freelance Developers
- Freelance Designers
- Project Managers

Secondary Users

- Agency Clients
- Development Team Members
- Workspace Administrators

User Roles

- Workspace Admin
- Project Manager
- Team Member / Developer
- Client

3. Core Problem

বর্তমানে client requirements বিভিন্ন জায়গায় ছড়িয়ে থাকে:

- Messenger
- WhatsApp
- Email

- Slack
- Phone Calls
- PDF
- Screenshots
- Documents
- Meetings

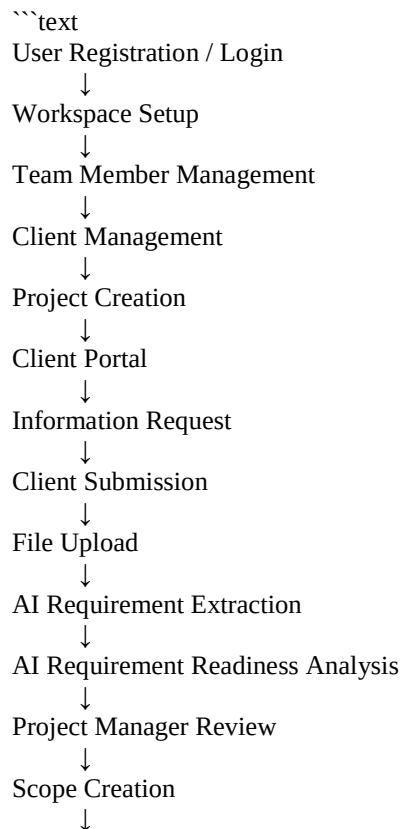
ফলে:

- Requirement হারিয়ে যেতে পারে
- Important information missing থাকতে পারে
- Client কী approve করেছে তা নিয়ে confusion হয়
- Informal request থেকে scope creep তৈরি হয়
- Developer এবং client-এর মধ্যে disagreement হতে পারে
- Extra work-এর জন্য agency payment নাও পেতে পারে
- Requirement থেকে development task তৈরি করতে সময় লাগে
- Project scope-এর পরিবর্তনের history properly track করা কঠিন হয়

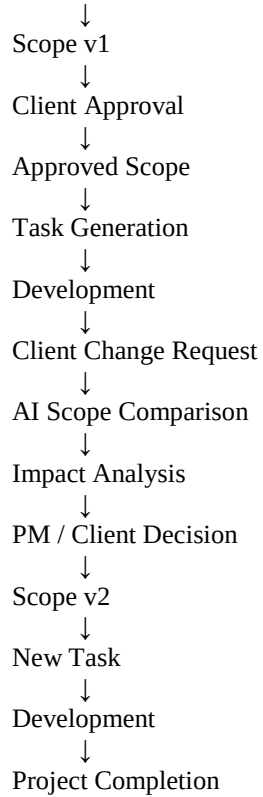
ScopeBridge এই informal communication এবং requirement workflow-কে একটি centralized এবং traceable system-এ রূপান্তর করবে।

4. Proposed Solution

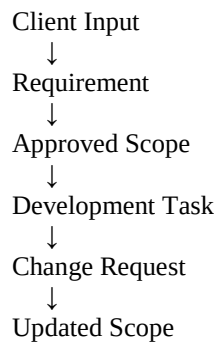
ScopeBridge-এর complete workflow হবে:



AI Acceptance Criteria Generation



Core Traceability



5. Core Functional Requirements

FR-01 — User Registration

নতুন user account তৈরি করতে পারবে।

Registration-এর জন্য:

- Name
- Email
- Password

প্রয়োজন হবে।

Registration সফল হলে user account তৈরি হবে।

FR-02 — User Login

Registered user email এবং password ব্যবহার করে login করতে পারবে।

Login সফল হলে:

Credentials
↓
Authentication
↓
JWT
↓
Dashboard

Secure authentication ব্যবহার করতে হবে।

FR-03 — Workspace Management

Workspace Admin একটি workspace তৈরি এবং manage করতে পারবে।

Workspace-এর মধ্যে থাকবে:

- Workspace Name
- Description
- Owner
- Members
- Clients
- Projects
- Settings

একটি workspace-এর data অন্য workspace থেকে isolated থাকবে।

FR-04 — Team Member Management

Workspace Admin team member invite করতে পারবে।

Admin পারবে:

- Member invite করতে
- Member role assign করতে
- Member remove করতে
- Member permission manage করতে

Roles

- Admin
 - Project Manager
 - Team Member
-

FR-05 — Client Management

Agency client manage করতে পারবে।

Client information:

- Name
- Email
- Company
- Phone
- Contact Information
- Status

Client-এর সাথে associated projects দেখা যাবে।

FR-06 — Project Management

Workspace-এর মধ্যে multiple project তৈরি করা যাবে।

Project-এর মধ্যে থাকবে:

- Project Name

- Description
- Client
- Team Members
- Start Date
- Deadline
- Status

Project-এর সাথে requirements, scope, tasks এবং change requests যুক্ত থাকবে।

FR-07 — Client Portal

Client-এর জন্য আলাদা portal থাকবে।

Client portal থেকে client পারবে:

- নিজের projects দেখতে
- Information Request দেখতে
- Information submit করতে
- File upload করতে
- Requirements দেখতে
- Scope দেখতে
- Scope approve/reject করতে
- Change Request submit করতে
- Project status দেখতে
- Notifications দেখতে

Client agency-এর internal data দেখতে পারবে না।

FR-08 — Information Request

Project Manager client-এর কাছ থেকে প্রয়োজনীয় information request করতে পারবে।

Example:

Please provide:

1. Target Audience
2. Product Categories
3. Payment Method
4. Delivery Process

- 5. Refund Policy
- 6. Branding Assets

প্রতিটি request-এর:

- Title
- Description
- Questions
- Deadline
- Status

থাকবে।

FR-09 — Client Submission

Client Information Request-এর উত্তর submit করতে পারবে।

Client:

- Text answer দিতে পারবে
- Form submit করতে পারবে
- File upload করতে পারবে
- Additional information দিতে পারবে

Submission project-এর সাথে linked থাকবে।

FR-10 — File Upload

Client project-related files upload করতে পারবে।

Supported file types:

- PDF
- DOC/DOCX
- Images
- Other approved project documents

File-এর সঙ্গে:

- File name

- File URL
- Uploaded by
- Project
- Upload date

সংরক্ষণ করতে হবে।

FR-11 — Requirement Management

Client submission থেকে structured requirements তৈরি করা যাবে।

Requirement-এর মধ্যে থাকবে:

- Title
- Description
- Category
- Priority
- Status
- Source
- Project
- Acceptance Criteria

Project Manager requirement:

- View
- Create
- Edit
- Delete
- Approve
- Request clarification

করতে পারবে।

FR-12 — AI Requirement Extraction

Client-এর submitted information এবং uploaded documents থেকে AI requirements extract করবে।

AI:

- Important information identify করবে
- Individual requirements তৈরি করবে
- Structured format-এ সাজাবে
- Requirement-এর source/context identify করবে

Example:

Client Document



Gemini AI



REQ-001 — User Registration

REQ-002 — Product Management

REQ-003 — Shopping Cart

REQ-004 — Online Payment

FR-13 — AI Requirement Readiness Analysis

AI requirement analyze করে identify করবে:

- Missing Information
- Ambiguous Requirements
- Contradictory Information
- Incomplete Requirements
- Unclear Requirements

Example:

Requirement Readiness: 72%

Missing:

- Payment Gateway
- Refund Process
- Delivery Rules

Project Manager client-এর কাছে clarification request করতে পারবে।

FR-14 — Scope Management

Ready requirements থেকে Project Manager project scope তৈরি করবে।

Scope-এর মধ্যে থাকবে:

- Features
- Deliverables
- Requirements
- Acceptance Criteria
- Exclusions
- Priority
- Deadline

FR-15 — AI Acceptance Criteria Generation

AI approved/ready requirement থেকে acceptance criteria-এর draft তৈরি করবে।

Example:

Requirement:
Online Payment

Given:
Customer has products in cart

When:
Customer completes checkout

Then:
Payment should be processed successfully
and an order should be created.

Project Manager AI-generated criteria review এবং edit করতে পারবে।

FR-16 — Scope Version Management

Scope-এর প্রতিটি approved version সংরক্ষণ করতে হবে।

Example:

Scope v1
↓
Client Approved
↓
Change Request
↓
Scope v2
↓
Client Approved

পুরোনো version delete হবে না।

প্রতিটি version-এর:

- Version Number
- Scope Data
- Created By
- Created At
- Status

থাকবে।

FR-17 — Scope Approval

Project Manager scope client-এর approval-এর জন্য পাঠাবে।

Client:

- Approve
- Reject
- Request Changes

করতে পারবে।

Approval history সংরক্ষণ করতে হবে।

FR-18 — Digital Approval

Client scope approve করলে একটি secure approval record তৈরি হবে।

Approval record:

- Scope Version
- Client
- Status
- Approved At
- Timestamp

সংরক্ষণ করবে।

FR-19 — Task Generation

Approved scope থেকে development task তৈরি করা যাবে।

Task-এর মধ্যে থাকবে:

- Title
- Description
- Assignee
- Priority
- Status
- Deadline
- Requirement
- Scope Version

Example:

Approved Scope
↓
Task Generation
↓
TASK-001 — Authentication API
TASK-002 — Login UI
TASK-003 — Product API
TASK-004 — Product UI
TASK-005 — Payment Integration

FR-20 — Task Management

Team Member assigned task manage করতে পারবে।

Task status:

Todo
↓
In Progress
↓
Review
↓
Completed

Project Manager task assign এবং monitor করতে পারবে।

FR-21 — Change Request Management

Development চলাকালীন client নতুন feature বা requirement চাইলে সেটি **Change Request** হিসেবে submit করতে হবে।

Example:

Client:

"We also want a Wishlist feature."

↓

Change Request:

Wishlist Feature

Change Request-এর মধ্যে থাকবে:

- Title
- Description
- Requested By
- Project
- Status
- Created At

FR-22 — Change Request Firewall

Client-এর Change Request সরাসরি development task হবে না।

Workflow:

Client Request

↓

Change Request

↓

AI Analysis

↓

Scope Comparison

↓

Impact Analysis

↓

Human Review

↓

Approval / Rejection

Approved না হওয়া পর্যন্ত Change Request development task হবে না।

FR-23 — AI Scope Change Analysis

AI নতুন Change Request এবং approved Scope compare করবে।

AI result:

IN SCOPE
OUT OF SCOPE
POSSIBLY RELATED

AI সম্ভাব্য impact identify করবে:

- Additional Development Work
- Database Changes
- API Changes
- UI Changes
- Timeline Impact

FR-24 — Change Request Approval

Project Manager Change Request review করবে।

প্রয়োজনে client approval নেওয়া হবে।

Possible status:

Pending
Under Review
Approved
Rejected
Implemented

FR-25 — Scope Update After Change

Approved Change Request-এর জন্য নতুন Scope Version তৈরি হবে।

Example:

Scope v1
+
Wishlist Change Request

↓
Scope v2

Scope v1 historical record হিসেবে থাকবে।

FR-26 — Project Communication

Project-specific communication system থাকবে।

Client এবং agency team project-related message exchange করতে পারবে।

Messages project-এর সাথে linked থাকবে।

FR-27 — Notifications

Users notification পাবে যখন:

- New Information Request
 - New Submission
 - File Uploaded
 - Requirement Updated
 - Scope Approval Requested
 - Scope Approved
 - Scope Rejected
 - Task Assigned
 - Change Request Submitted
 - Change Request Approved
 - Change Request Rejected
 - Deadline Approaching
-

FR-28 — Automated Email Reminder

Client required information অথবা file submit না করলে system reminder পাঠাতে পারবে।

Background job ব্যবহার করে automated reminders পরিচালনা করা হবে।

FR-29 — Activity / Audit Trail

Important system activities record করতে হবে।

Example:

Project Created
Client Invited
Information Request Sent
Client Submitted Information
AI Analysis Completed
Requirement Updated
Scope v1 Created
Scope v1 Approved
Task Created
Task Completed
Change Request Submitted
Change Request Approved
Scope v2 Created

প্রতিটি activity-এর সঙ্গে:

- User
- Action
- Entity
- Timestamp

থাকবে।

FR-30 — Dashboard & Analytics

Workspace Admin এবং Project Manager project-related analytics দেখতে পারবে।

Dashboard-এ:

- Total Projects
- Active Projects
- Total Clients
- Pending Approvals
- Total Requirements
- Total Tasks
- Completed Tasks
- Pending Change Requests
- Approved Changes
- Rejected Changes

দেখানো হবে।

6. AI Requirements

ScopeBridge-এ minimum 5টি **meaningful AI feature** থাকতে হবে।

AI-01 — AI Checklist Generator

Project type অনুযায়ী client-এর কাছ থেকে কী কী information প্রয়োজন AI suggest করবে।

Example:

E-commerce Project

Required:

- ✓ Business Information
 - ✓ Product Information
 - ✓ Branding Assets
 - ✓ Payment Requirements
 - ✓ Delivery Information
 - ✓ Refund Policy
-

AI-02 — AI Requirement Extraction

Client document, submission অথবা text থেকে AI structured requirements extract করবে।

AI-03 — AI Requirement Readiness Analysis

AI identify করবে:

- Missing Information
 - Ambiguous Requirements
 - Contradictions
 - Incomplete Requirements
-

AI-04 — AI Acceptance Criteria Generator

Requirements থেকে structured acceptance criteria generate করবে।

Project Manager final criteria review করবে।

AI-05 — AI Scope Change Analyzer

New Change Request এবং approved scope compare করবে।

Result:

In Scope
Out of Scope
Possibly Related

এবং সম্ভাব্য impact দেখাবে।

AI Failure Handling

AI service unavailable হলে application crash করবে না।

System manual workflow allow করবে:

AI Failed
↓
Manual Mode
↓
User manually creates/edits data

AI Decision Rule

AI কোনো final approval decision নেবে না।

AI শুধু:

- Analyze
- Extract
- Suggest
- Compare
- Generate Draft

করবে।

Final decision থাকবে:

Project Manager / Agency / Client-এর হাতে।

7. Non-Functional Requirements

Performance

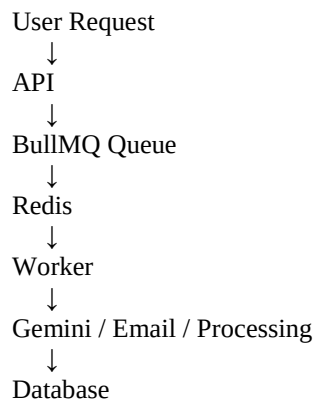
- API response reasonably fast হতে হবে
- Unnecessary database queries avoid করতে হবে
- Pagination ব্যবহার করতে হবে
- Large data-এর জন্য optimized queries ব্যবহার করতে হবে
- Heavy AI operations background processing-এ চলবে

Background Processing

Redis + BullMQ ব্যবহার করে:

- AI processing
- File processing
- Email reminders
- Long-running jobs

background-এ চালানো হবে।



Security

System-এ থাকতে হবে:

- Password hashing
- JWT Authentication
- Role-Based Authorization
- Protected APIs
- Environment Variables
- Input Validation
- File Validation
- Rate Limiting
- Workspace Data Isolation

Reliability

- Proper API error handling
- Database error handling
- AI failure handling
- Background job failure handling
- File upload failure handling
- Graceful fallback

Scalability

Future-এ system যেন support করতে পারে:

- More Workspaces
- More Users
- More Clients
- More Projects
- More Requirements
- More AI Features

Responsiveness

Application properly কাজ করতে হবে:

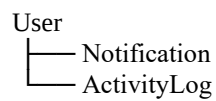
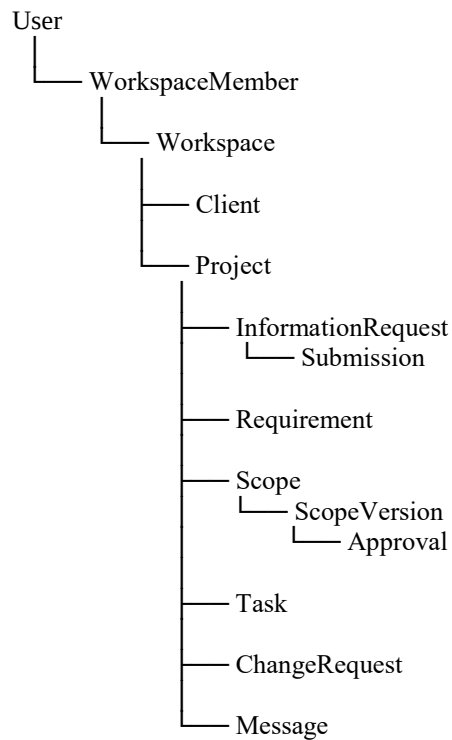
- Desktop
- Tablet
- Mobile

8. Database Requirements

Core database entities:

User
Workspace
WorkspaceMember
Client
Project
InformationRequest
Submission
Requirement
Scope
ScopeVersion
Approval
Task
ChangeRequest
Message
Notification
ActivityLog

Main Relationships



প্রতিটি entity-এর জন্য define করতে হবে:

- Primary Key
- Foreign Key
- Required Fields

- Relationships
 - Constraints
 - Indexes যেখানে প্রয়োজন
-

9. API Requirements

Backend REST API provide করবে।

Authentication

POST /api/auth/register
POST /api/auth/login
POST /api/auth/logout
POST /api/auth/refresh
POST /api/auth/forgot-password
POST /api/auth/reset-password

Users

GET /api/users
GET /api/users/:id
PATCH /api/users/:id

Workspaces

POST /api/workspaces
GET /api/workspaces
GET /api/workspaces/:id
PATCH /api/workspaces/:id
DELETE /api/workspaces/:id

Workspace Members

POST /api/workspaces/:id/members
GET /api/workspaces/:id/members
PATCH /api/workspaces/:id/members/:memberId
DELETE /api/workspaces/:id/members/:memberId

Clients

POST /api/clients
GET /api/clients
GET /api/clients/:id
PATCH /api/clients/:id

DELETE /api/clients/:id

Projects

POST /api/projects
GET /api/projects
GET /api/projects/:id
PATCH /api/projects/:id
DELETE /api/projects/:id

Information Requests

POST /api/information-requests
GET /api/information-requests
GET /api/information-requests/:id
PATCH /api/information-requests/:id

Submissions

POST /api/submissions
GET /api/submissions
GET /api/submissions/:id

Requirements

POST /api/requirements
GET /api/requirements
GET /api/requirements/:id
PATCH /api/requirements/:id
DELETE /api/requirements/:id

Scopes

POST /api/scopes
GET /api/scopes/:id
POST /api/scopes/:id/versions
GET /api/scopes/:id/versions
GET /api/scope-versions/:id

Approvals

POST /api/approvals
GET /api/approvals
POST /api/approvals/:id/approve
POST /api/approvals/:id/reject

Tasks

POST /api/tasks
GET /api/tasks
GET /api/tasks/:id
PATCH /api/tasks/:id
DELETE /api/tasks/:id

Change Requests

POST /api/change-requests
GET /api/change-requests
GET /api/change-requests/:id
PATCH /api/change-requests/:id
POST /api/change-requests/:id/analyze
POST /api/change-requests/:id/approve
POST /api/change-requests/:id/reject

Messages

POST /api/messages
GET /api/messages

Notifications

GET /api/notifications
PATCH /api/notifications/:id/read

AI

POST /api/ai/checklist
POST /api/ai/extract
POST /api/ai/readiness
POST /api/ai/acceptance-criteria
POST /api/ai/scope-analysis

প্রতিটি API-তে প্রয়োজন অনুযায়ী:

- Authentication
- Authorization
- Validation
- Error Handling
- Proper HTTP Status Codes
- Consistent Response Format

থাকবে।

10. Technology Requirements

Frontend

- Next.js
- TypeScript
- Tailwind CSS
- shadcn/ui

Backend

- Node.js
- Express.js
- TypeScript

Database

- PostgreSQL
- Prisma ORM

Authentication

- JWT
- Argon2

Validation

- Zod

AI

- Gemini API

Real-Time Communication

- Socket.IO

Background Jobs

- Redis
- BullMQ

File Storage

- Cloudinary / S3-compatible Storage

API Documentation

- Swagger / OpenAPI

Testing

- Playwright

Deployment

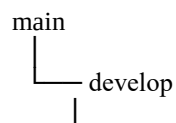
- Vercel
 - Render
 - Neon PostgreSQL
-

11. Development Requirements

Team অবশ্যই:

- GitHub ব্যবহার করবে
- main branch maintain করবে
- develop branch maintain করবে
- Feature branch ব্যবহার করবে
- Pull Request তৈরি করবে
- Code Review করবে
- Meaningful Commit করবে
- Issue Tracking করবে
- Project Management Board maintain করবে
- Regular Integration করবে
- Bugs Track করবে

Suggested Branch Structure



- feature/auth
- feature/workspace
- feature/client
- feature/project
- feature/requirements
- feature/scope
- feature/task
- feature/ai
- feature/change-request

12. Documentation Requirements

Final project-এর সঙ্গে থাকতে হবে:

1. Project Overview
 2. Problem Statement
 3. Target Users
 4. Proposed Solution
 5. Core Features
 6. AI Features
 7. Technology Stack
 8. ERD
 9. Database Schema
 10. System Architecture
 11. SDA Workflow
 12. API Structure
 13. API Documentation
 14. Git Workflow
 15. Testing Strategy
 16. Setup Instructions
 17. Environment Configuration
 18. Deployment Information
-

13. MVP Success Criteria

Final MVP complete ধরা হবে যখন:

Core Product

- ☐ User Registration & Login
- ☐ Workspace Management

- ☐ Team Member Management
- ☐ RBAC
- ☐ Client Management
- ☐ Project Management
- ☐ Client Portal
- ☐ Information Request
- ☐ Client Submission
- ☐ File Upload
- ☐ Requirement Management
- ☐ Scope Management
- ☐ Scope Versioning
- ☐ Client Approval
- ☐ Task Generation
- ☐ Task Management
- ☐ Change Request Firewall
- ☐ Project Communication
- ☐ Notifications
- ☐ Audit Trail
- ☐ Dashboard & Analytics

AI

- ☐ AI Checklist Generator
- ☐ AI Requirement Extraction
- ☐ AI Requirement Readiness Analysis
- ☐ AI Acceptance Criteria Generator
- ☐ AI Scope Change Analyzer
- ☐ Gemini API Integration
- ☐ AI Failure Handling
- ☐ Human-controlled Final Decision

Engineering

- ☐ REST API
- ☐ Validation
- ☐ Error Handling
- ☐ JWT Authentication
- ☐ RBAC
- ☐ PostgreSQL
- ☐ Prisma ORM
- ☐ Redis
- ☐ BullMQ
- ☐ Socket.IO
- ☐ Git/GitHub Workflow
- ☐ Pull Requests
- ☐ Code Review
- ☐ Playwright Testing

Architecture

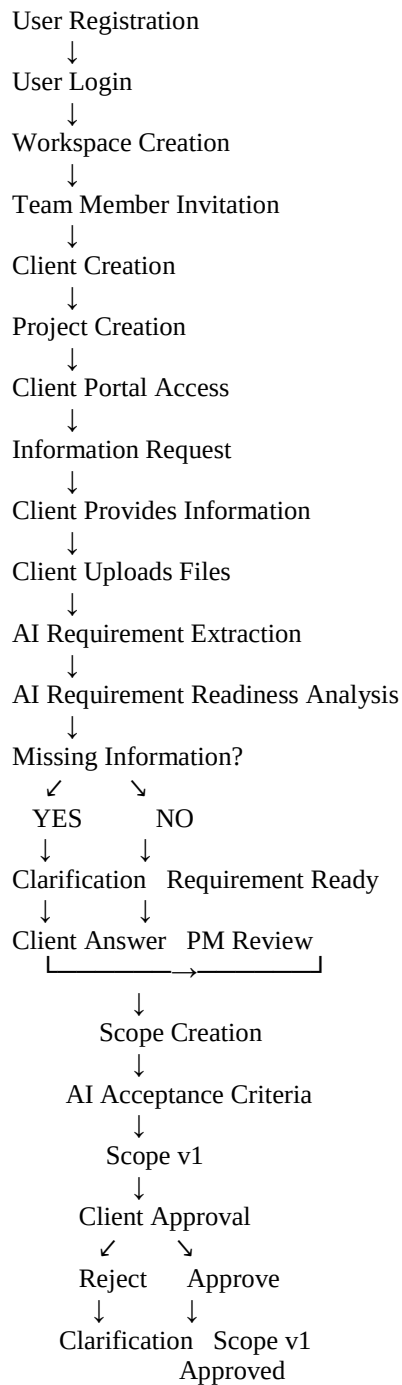
- ☐ ERD
- ☐ Database Schema
- ☐ System Architecture
- ☐ SDA Workflow
- ☐ API Structure

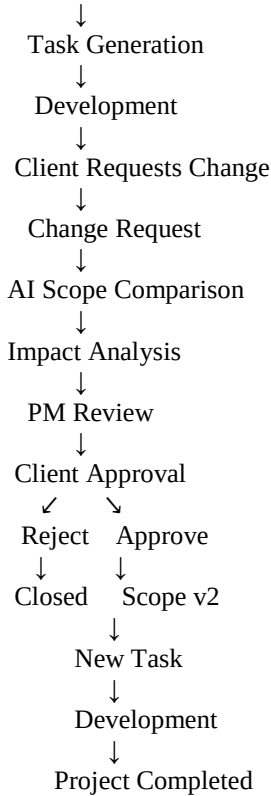
Deployment

- ☐ Frontend Deployed
- ☐ Backend Deployed
- ☐ PostgreSQL Connected
- ☐ Redis Connected
- ☐ Background Worker Deployed
- ☐ Public URL
- ☐ No Dummy Data
- ☐ All Major Routes Functional

- ☐ All Major Forms Functional
- ☐ Responsive UI

14. Complete Project Workflow





15. One-Line Project Requirement

ScopeBridge must provide a complete and traceable client-to-development workflow that transforms client input into structured requirements, AI-assisted analysis, approved version-controlled project scope, development tasks, and controlled scope changes through a formal Change Request Firewall.

Core idea

****ScopeBridge is not simply a project management or task management system.****

এর মূল workflow হলো:

```

``text
Client
↓
Requirement
↓
AI Analysis
↓
Approved Scope
↓
Developer Task
↓
  
```

Development
↓
Change Request
↓
AI Scope Analysis
↓
Approval
↓
New Scope Version

এটিই project-এর **main business logic এবং USP**। Strategic plan-এও এই traceable scope এবং Change Request Firewall-কে core differentiator হিসেবে নির্ধারণ করা হয়েছে। Pasted markdown